

Lovisha Bansal

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Education

Thapar Institute of Engineering and Technology

B.Tech in Computer Science and Engineering

CGPA: 8.71

Patiala, Punjab

Aug 2023 – June 2027

SD Senior Secondary School

Senior Secondary (PCM) – Score: 95.60%

Sri Ganganagar, Rajasthan

May 2021 – April 2022

Relevant Coursework:

Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems, Computer Networks, Design and Analysis of Algorithms, Computer Organization and Architecture (COA).

Projects

AI-Powered Fashion Recommendation System | *Python, Stable Diffusion, Deep Learning*

- Built a recommendation engine generating personalized outfit suggestions using image-to-image models.
- Implemented virtual try-on mechanics using conditional generative networks for realistic garment mapping.
- Integrated an interactive chatbot interface to provide user assistance and handle tailored apparel queries.
- Managed end-to-end ML workflow including high-dimensional dataset preprocessing, training, and evaluation.

Disease Prediction Framework | *Python, Machine Learning, Scikit-learn*

- Engineered predictive models for diagnostic evaluation of heart, diabetes, and kidney chronic conditions.
- Conducted extensive EDA and statistical feature selection on clinical data to minimize model variance.
- Deployed and benchmarked Logistic Regression, Random Forest, Decision Tree, and KNN classification matrices.
- Wrapped models into an end-to-end pipeline architecture suitable for scalable healthcare applications.

Self-Balancing Robot | *Arduino, Embedded Systems, PID Control*

- Developed a 2.1 kg dual-wheel self-balancing robot managing automated real-time inertial feedback.
- Wired MPU6050 accelerometer/gyroscope, NEMA 17 stepper motors, and TB6600 drivers with an Arduino board.
- Tuned PID control loops to dramatically improve active equilibrium response times and angular stability.

Technical Achievements

- **1st Runner-Up** out of multiple competing teams in the Capture The Flag (CTF) Cybersecurity Event, TIET Colloquium 2025 [Certificate](#)
- **Self-Balancing Robot Certification:** Certified technical proficiency in embedded hardware validation [Certificate](#)
- **Global Coding Contests:** Actively participating in LeetCode Weekly and Biweekly contests

Skills

Programming Languages: C++, Python, SQL, R, MATLAB, HTML/CSS, JavaScript (Basics)

Machine Learning & Data Science: Supervised Learning, Deep Learning (CNNs), Exploratory Data Analysis (EDA), Feature Engineering, Model Optimization, Data Preprocessing

Libraries & Frameworks: NumPy, Pandas, Scikit-Learn, Matplotlib, Seaborn

Developer Tools & Databases: Git, GitHub, VS Code, Jupyter Notebook, MySQL, Arduino IDE

Core Concepts: Machine Learning, Model Evaluation, Problem Solving

Languages: English, Hindi, Punjabi